

TF-blood_vs_primary

2026-06-15

Purpose

This exploratory analysis compares matched blood and primary-tumor telomere fusion rates. It separates signal presence from signal magnitude because both variables are strongly zero-inflated.

Setup and data loading

```
library(tidyverse)
library(here)
data <- readxl::read_xlsx(here('data', 'processed', 'PCAWG_primary.xlsx'))
clean_data <- data
summary(clean_data)
```

donor_id	cancer_type	alt_status	tf_primary_rate
Length:2014	Length:2014	Length:2014	Min. : 0.00000
Class :character	Class :character	Class :character	1st Qu.: 0.00000
Mode :character	Mode :character	Mode :character	Median : 0.03931
			Mean : 0.13568
			3rd Qu.: 0.10251
			Max. :19.73776

tf_blood_rate	telomere_insertion_rate	telomere_content_log2
Min. :0.00000	Min. :0.00000	Min. : -4.00250
1st Qu.:0.00000	1st Qu.:0.00000	1st Qu.: -0.92876
Median :0.02628	Median :0.00000	Median : -0.52968
Mean :0.04660	Mean :0.03307	Mean : -0.43589

3rd Qu.:0.05967	3rd Qu.:0.02195	3rd Qu.: -0.07054
Max. :1.53097	Max. :5.38979	Max. : 3.84469

TERT_FPKM	ATAGGG_singleton_dist	CTAGGG_singleton_dist
Min. : 0.00000	Min. : -4.35757	Min. : -3.49674
1st Qu.: 0.02069	1st Qu.: -0.68156	1st Qu.: -0.38418
Median : 0.10630	Median : -0.03350	Median : 0.05423
Mean : 0.90171	Mean : -0.06042	Mean : 0.06588
3rd Qu.: 0.37979	3rd Qu.: 0.57645	3rd Qu.: 0.53662
Max. :53.76790	Max. : 4.47470	Max. : 7.80772
NA's :1166	NA's :28	NA's :2
GTAGGG_singleton_dist	TAAGGG_singleton_dist	TCAGGG_singleton_dist
Min. : -2.981384	Min. : -4.67195	Min. : -4.27208
1st Qu.: -0.359368	1st Qu.: -0.53849	1st Qu.: -0.56813
Median : 0.017444	Median : 0.08092	Median : 0.02901
Mean : 0.001582	Mean : 0.06257	Mean : 0.08148
3rd Qu.: 0.352901	3rd Qu.: 0.67706	3rd Qu.: 0.70046
Max. : 4.253404	Max. : 3.87943	Max. : 7.64722
	NA's :28	NA's :18
TGAGGG_singleton_dist	TTCGGG_singleton_dist	TTGGGG_singleton_dist
Min. : -5.27154	Min. : -3.848796	Min. : -3.109601
1st Qu.: -0.37005	1st Qu.: -0.568133	1st Qu.: -0.417046
Median : 0.06673	Median : 0.007297	Median : -0.019229
Mean : 0.07103	Mean : 0.102063	Mean : 0.002303
3rd Qu.: 0.46860	3rd Qu.: 0.573244	3rd Qu.: 0.429475
Max. : 6.67267	Max. : 8.739318	Max. : 4.584889
	NA's :5	NA's :2
TTTGGG_singleton_dist	Specimen.Type.Summary	cancer_group
Min. : -4.1941	Length:2014	Length:2014
1st Qu.: -0.5610	Class :character	Class :character
Median : -0.1030	Mode :character	Mode :character
Mean : -0.1309		
3rd Qu.: 0.3465		
Max. : 2.7484		

Marginal distributions

Summary and zero prevalence

```
summary(clean_data$tf_blood_rate)
```

```
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.02628 0.04660 0.05967 1.53097
```

```
summary(clean_data$tf_primary_rate)
```

```
      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00000 0.00000 0.03931 0.13568 0.10251 19.73776
```

```
zero_blood <- sum(clean_data$tf_blood_rate == 0)/nrow(clean_data) * 100
zero_primary <- sum(clean_data$tf_primary_rate == 0)/nrow(clean_data) * 100

cat('Percentage of zeros in tf_blood_rate: ', zero_blood, "%\n")
```

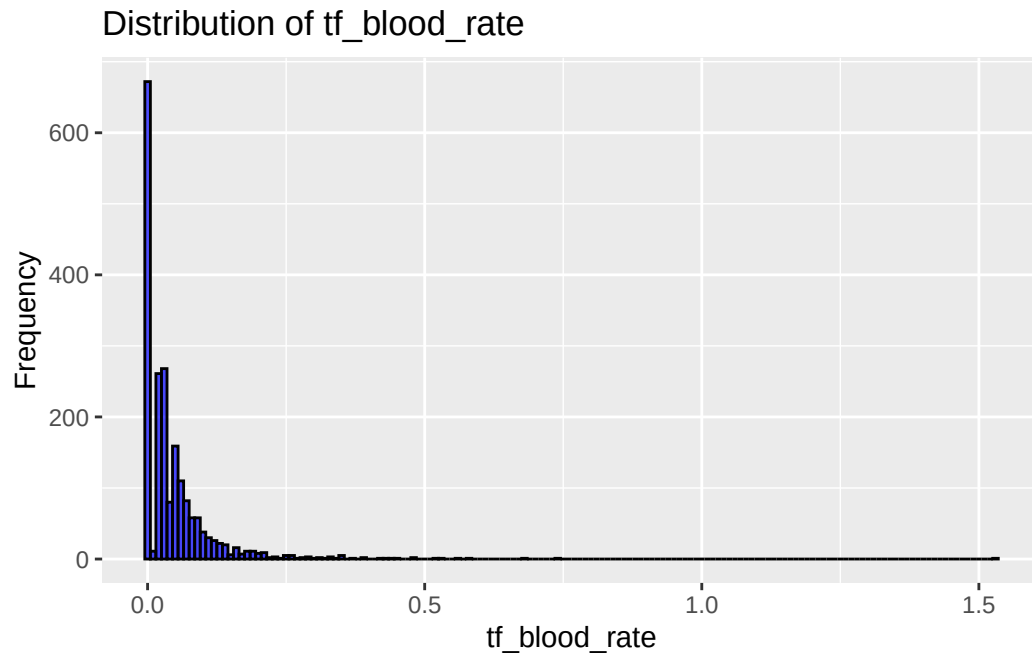
```
Percentage of zeros in tf_blood_rate:  33.36643 %
```

```
cat('Percentage of zeros in tf_primary_rate: ', zero_primary, "%\n")
```

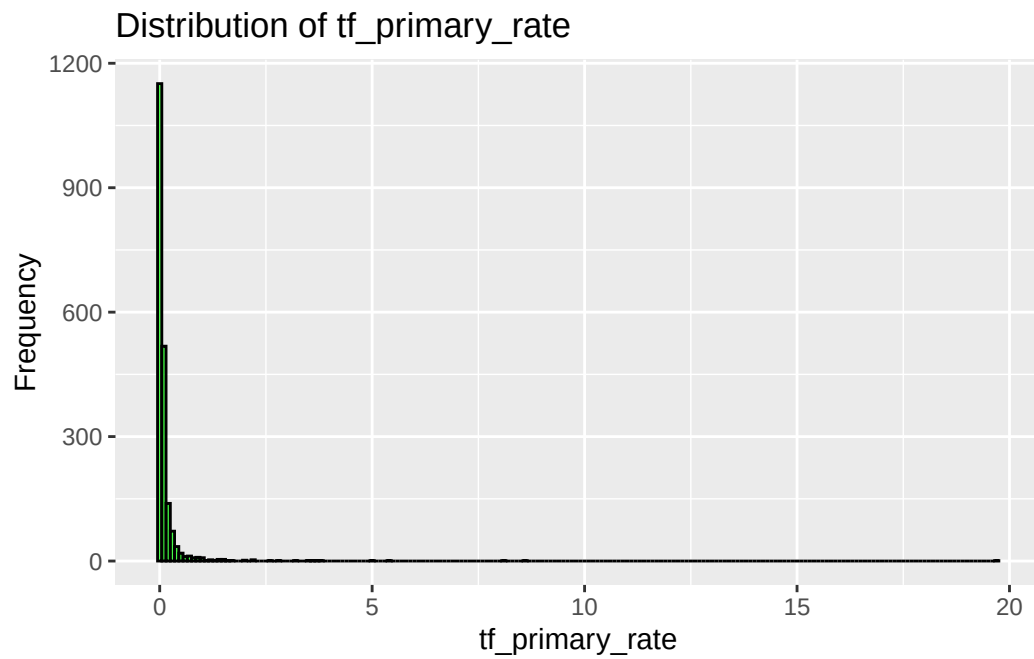
```
Percentage of zeros in tf_primary_rate:  27.95432 %
```

Distribution plots

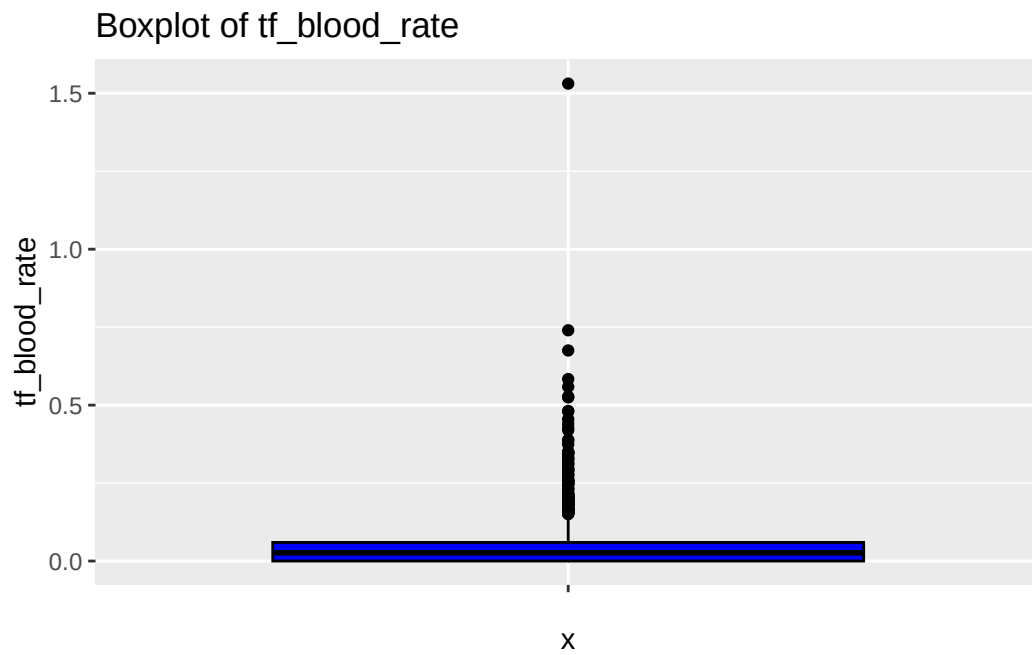
```
clean_data %>%
  ggplot(aes(x = tf_blood_rate)) +
  geom_histogram(binwidth = 0.01, fill = 'blue', color = 'black', alpha = 0.7) +
  labs(title = 'Distribution of tf_blood_rate', x = 'tf_blood_rate', y = 'Frequency')
```



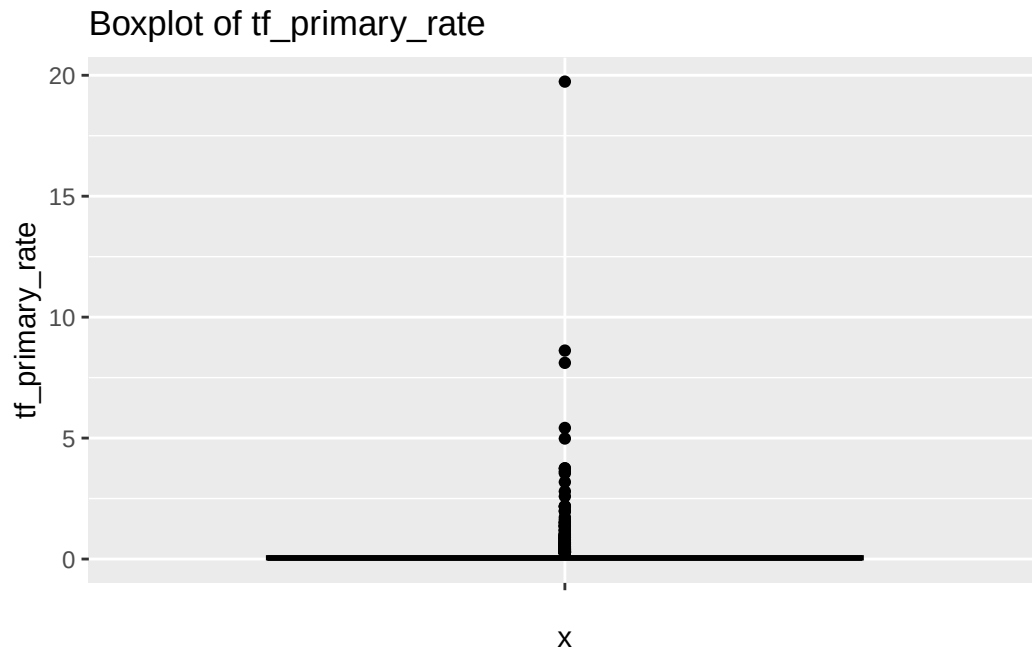
```
clean_data %>%  
  ggplot(aes(x = tf_primary_rate)) +  
  geom_histogram(binwidth = 0.1, fill = 'green', color = 'black', alpha = 0.7) +  
  labs(title = 'Distribution of tf_primary_rate', x = 'tf_primary_rate', y = 'Frequency')
```



```
clean_data %>%  
  ggplot(aes(x = '', y = tf_blood_rate)) +  
  geom_boxplot(fill = 'blue', color = 'black') +  
  labs(title = 'Boxplot of tf_blood_rate', y = 'tf_blood_rate')
```



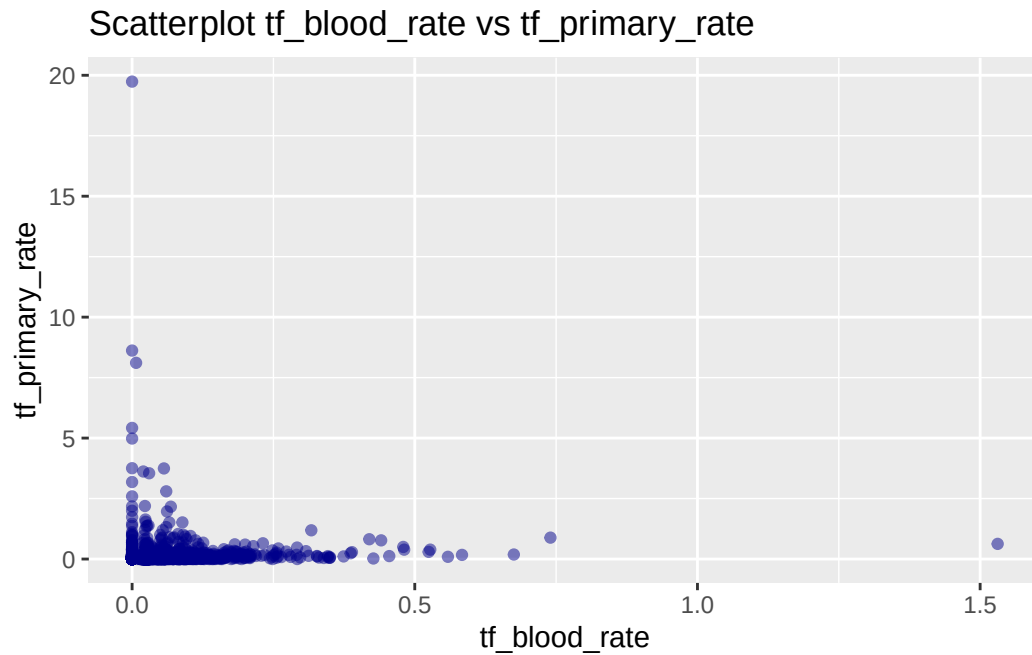
```
clean_data %>%  
  ggplot(aes(x = '', y = tf_primary_rate)) +  
  geom_boxplot(fill = 'green', color = 'black') +  
  labs(title = 'Boxplot of tf_primary_rate', y = 'tf_primary_rate')
```



Joint distributions

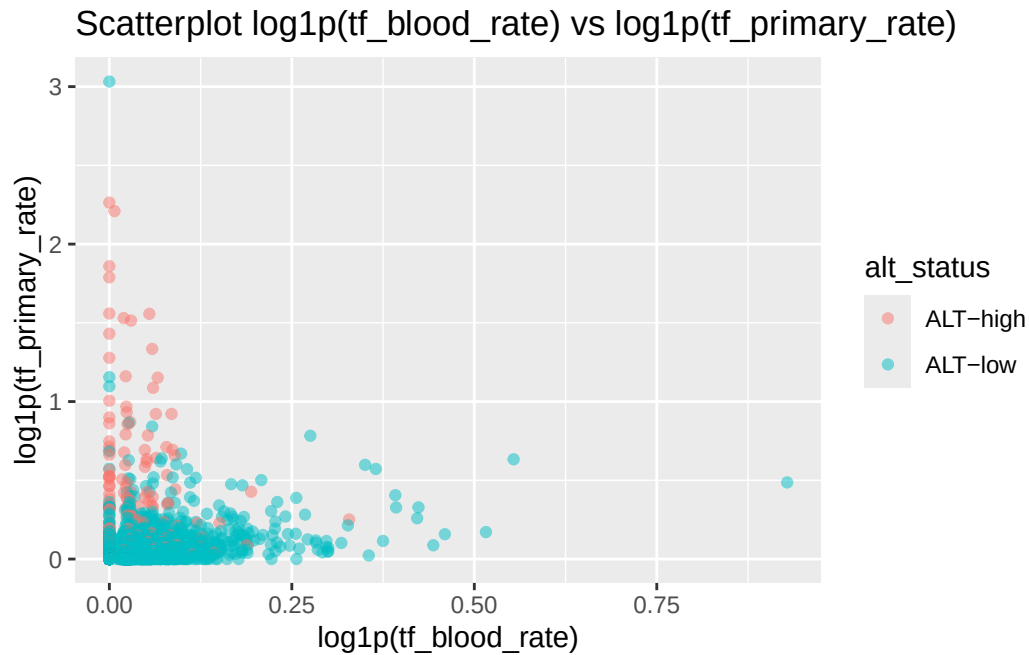
Scatterplot raw

```
clean_data %>%  
  ggplot(aes(tf_blood_rate, tf_primary_rate)) +  
  geom_point(alpha = 0.5, color = 'darkblue') +  
  labs(title = 'Scatterplot tf_blood_rate vs tf_primary_rate', x = 'tf_blood_rate', y = 'tf_
```



Scatterplot log1p transformed

```
clean_data %>%  
  ggplot(aes(log1p(tf_blood_rate), log1p(tf_primary_rate), color = alt_status)) +  
  geom_point(alpha = 0.5) +  
  labs(title = 'Scatterplot log1p(tf_blood_rate) vs log1p(tf_primary_rate)', x = 'log1p(tf_b
```

Signal presence: zero versus positive

The binary comparison asks whether detectable fusion signal tends to occur in both sample sources, independently of the magnitude among positive samples.

```
clean_data <- clean_data %>%
  mutate(blood_pos = ifelse(tf_blood_rate > 0, 1, 0),
         primary_pos = ifelse(tf_primary_rate > 0, 1, 0))

table(clean_data$blood_pos, clean_data$primary_pos, dnn = c('blood_pos', 'primary_pos'))
```

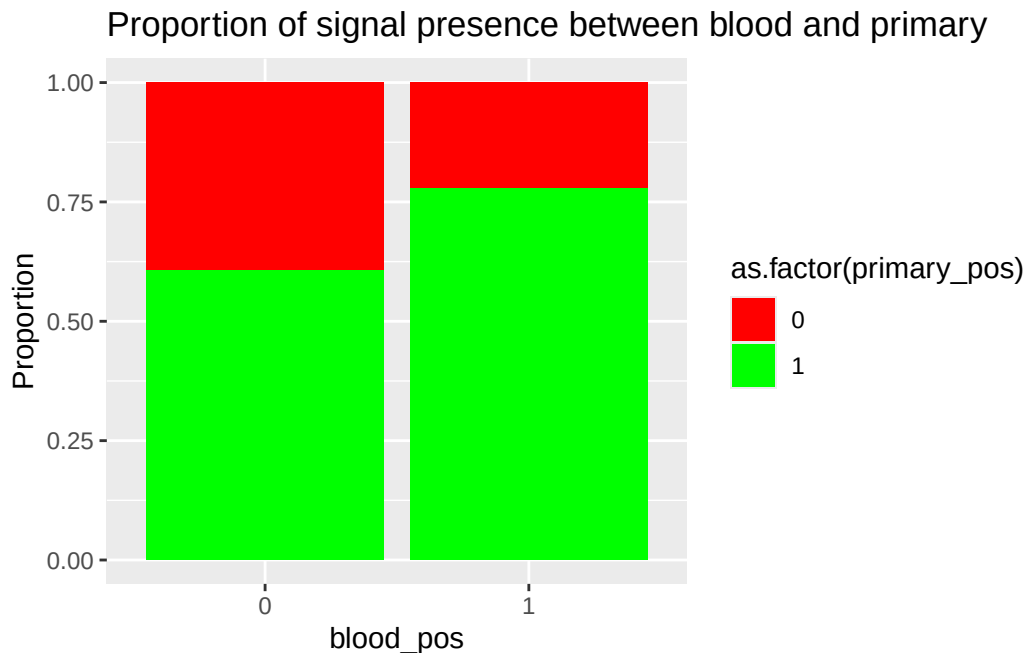
	primary_pos	
blood_pos	0	1
0	264	408
1	299	1043

```
chisq.test(table(clean_data$blood_pos, clean_data$primary_pos))
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: table(clean_data$blood_pos, clean_data$primary_pos)
X-squared = 63.455, df = 1, p-value = 1.641e-15
```

```
ggplot(clean_data, aes(x = as.factor(blood_pos), fill = as.factor(primary_pos))) +
  geom_bar(position = 'fill') +
  labs(title = 'Proportion of signal presence between blood and primary', x = 'blood_pos', y = 'Proportion') +
  scale_fill_manual(values = c('red', 'green'), labels = c('0', '1'))
```



Discordant samples: primary TF = 0, blood TF > 0

Definition and ALT status enrichment

```
# Samples with no detectable primary TF but positive blood TF
discordant <- clean_data %>%
  mutate(group = case_when(
    primary_pos == 0 & blood_pos == 1 ~ 'primary=0, blood>0',
    primary_pos == 1 & blood_pos == 1 ~ 'both>0',
    primary_pos == 1 & blood_pos == 0 ~ 'primary>0, blood=0',
    TRUE ~ 'both=0'
```

```
))

cat("Group counts:\n")
```

Group counts:

```
print(table(discordant$group))
```

```

      both=0      both>0 primary=0, blood>0 primary>0, blood=0
      264      1043      299      408

```

```

# Focus on discordant group vs the rest (excluding both=0 as uninformative)
disc_subset <- discordant %>%
  filter(group %in% c('primary=0, blood>0', 'both>0', 'primary>0, blood=0')) %>%
  mutate(is_discordant = group == 'primary=0, blood>0')

cat("\nALT status in discordant group (primary=0, blood>0):\n")

```

ALT status in discordant group (primary=0, blood>0):

```
print(table(discordant$alt_status[discordant$group == 'primary=0, blood>0']))
```

```

ALT-high  ALT-low
      9      290

```

```
cat("\nALT status in non-discordant groups:\n")
```

ALT status in non-discordant groups:

```
print(table(discordant$alt_status[discordant$group != 'primary=0, blood>0']))
```

```

ALT-high  ALT-low
     161     1554

```

```
# Fisher's exact test: is ALT-low enriched in discordant samples?
ct <- table(disc_subset$is_discordant, disc_subset$alt_status)
rownames(ct) <- c('other', 'discordant')
print(ct)
```

	ALT-high	ALT-low
other	150	1301
discordant	9	290

```
fisher_res <- fisher.test(ct)
cat("\nFisher's exact test (discordant vs other groups, ALT-low enrichment):\n")
```

Fisher's exact test (discordant vs other groups, ALT-low enrichment):

```
print(fisher_res)
```

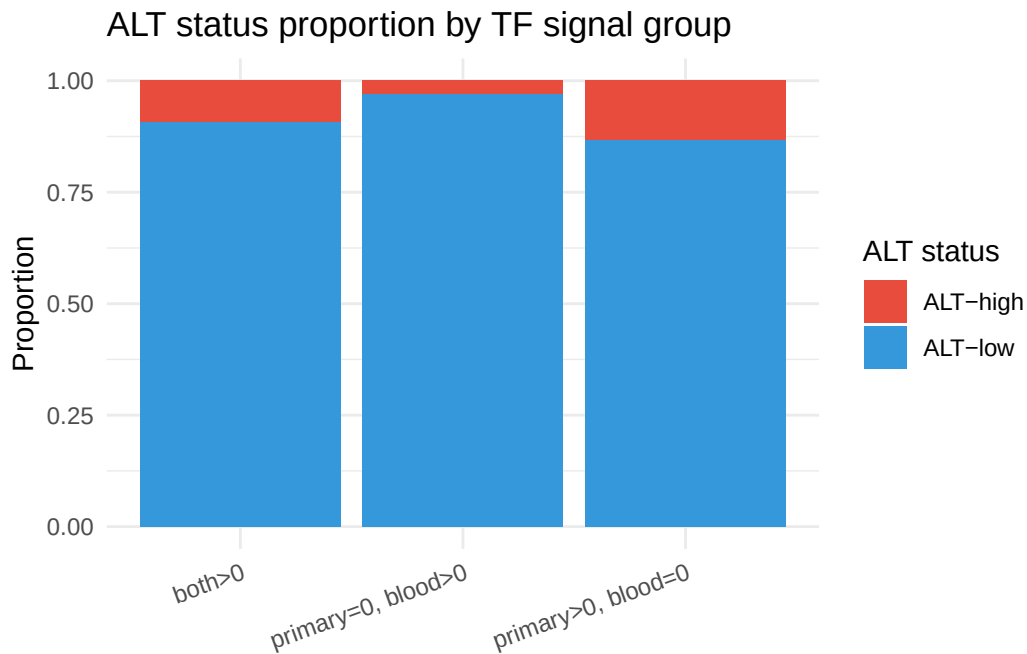
Fisher's Exact Test for Count Data

```
data: ct
p-value = 1.261e-05
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 1.874170 8.376733
sample estimates:
odds ratio
 3.713292
```

ALT status proportion plot

```
# Bar chart: ALT status proportions by group
discordant %>%
  filter(group != 'both=0') %>%
  ggplot(aes(x = group, fill = alt_status)) +
  geom_bar(position = 'fill') +
  scale_fill_manual(values = c('ALT-high' = '#E74C3C', 'ALT-low' = '#3498DB')) +
```

```
labs(
  title = 'ALT status proportion by TF signal group',
  x = NULL, y = 'Proportion', fill = 'ALT status'
) +
theme_minimal() +
theme(axis.text.x = element_text(angle = 20, hjust = 1))
```



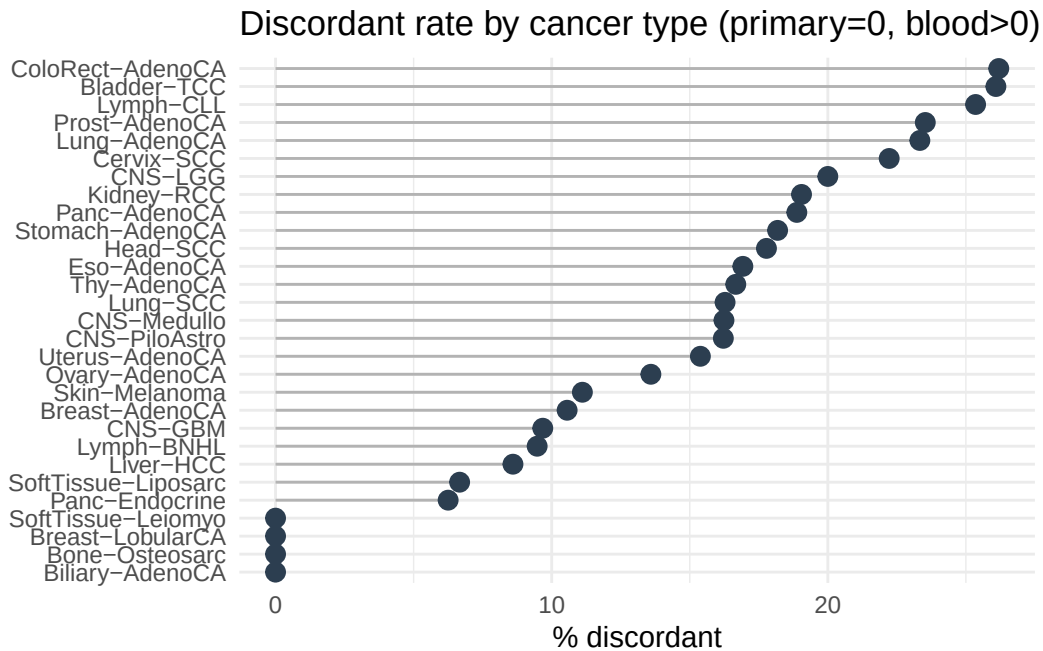
Discordant rate by cancer type

```
disc_pct <- discordant %>%
  group_by(cancer_type) %>%
  summarise(
    n_total = n(),
    n_disc = sum(group == 'primary=0, blood>0'),
    pct_disc = n_disc / n_total * 100,
    .groups = 'drop'
  ) %>%
  filter(n_total >= 10) %>%
  arrange(desc(pct_disc))

print(disc_pct)
```

```
# A tibble: 29 x 4
  cancer_type      n_total n_disc pct_disc
  <chr>          <int>  <int>   <dbl>
1 ColoRect-AdenoCA      42    11    26.2
2 Bladder-TCC           23     6    26.1
3 Lymph-CLL             71    18    25.4
4 Prost-AdenoCA        136    32    23.5
5 Lung-AdenoCA          30     7    23.3
6 Cervix-SCC            18     4    22.2
7 CNS-LGG               15     3     20
8 Kidney-RCC           126    24    19.0
9 Panc-AdenoCA          196    37    18.9
10 Stomach-AdenoCA       55    10    18.2
# i 19 more rows
```

```
ggplot(disc_pct, aes(x = pct_disc, y = reorder(cancer_type, pct_disc))) +
  geom_segment(aes(x = 0, xend = pct_disc,
                  y = reorder(cancer_type, pct_disc),
                  yend = reorder(cancer_type, pct_disc)),
              color = 'grey70') +
  geom_point(size = 3, color = '#2C3E50') +
  labs(
    title = 'Discordant rate by cancer type (primary=0, blood>0)',
    x = '% discordant', y = NULL
  ) +
  theme_minimal()
```



Correlation among jointly positive samples

Restricting to samples with positive signal in both sources avoids letting the shared mass at zero dominate the rank correlation.

```
positive_data <- clean_data %>%
  filter(blood_pos == 1 & primary_pos == 1)

cor.test(positive_data$tf_blood_rate[positive_data$blood_pos == 1],
  positive_data$tf_primary_rate[positive_data$primary_pos == 1],
  method = 'spearman')
```

Spearman's rank correlation rho

```
data: positive_data$tf_blood_rate[positive_data$blood_pos == 1] and positive_data$tf_primary_rate[positive_data$primary_pos == 1]
S = 144483840, p-value = 1.401e-14
alternative hypothesis: true rho is not equal to 0
sample estimates:
rho
0.2359567
```

Relationship among jointly positive samples

```
ggplot(positive_data, aes(x = log1p(tf_blood_rate), y = log1p(tf_primary_rate))) +  
  geom_point(alpha = 0.5, color = 'darkred') +  
  geom_smooth(method = 'loess', color = 'blue') +  
  labs(title = 'tf_blood_rate vs tf_primary_rate', x = 'log1p(tf_blood_rate)', y = 'log1p(tf
```

